CLASS 10 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The kidneys in human beings are a part of the system for: 1

(a) excretion **(b)** respiration
(c) transportation of oxygen **(d)** nutrition

2. A part of the neuron where information is acquired is the: 1

(a) axon **(b)** cell body **(c)** axon end **(d)** dendrite

3. Binary fission in Amoeba differs from that in Leishmania because in Leishmania the division occurs 1

(a) in the nucleus alone
(b) at a definite plane related to the whip-like flagellum
(c) in any plane
(d) only after sporulation

4. Who is called the Father of Genetics? 1

(a) Gregor Johann Mendel **(b)** Robert Hooke
(c) Charles Darwin **(d)** Louis Pasteur

5. Aerobic respiration in humans yields: 1

(a) CO_2, water and energy **(b)** ethanol and CO_2
(c) only energy **(d)** lactic acid

6. The movement of a shoot towards light is called: 1

(a) hydrotropism **(b)** chemotropism **(c)** phototropism **(d)** geotropism

7. The male gamete in a flowering plant is formed in the 1

(a) pollen grain **(b)** ovary **(c)** stigma **(d)** ovule

8. In a dihybrid cross between round yellow (RRYY) and wrinkled green (rryy) seeds, the F1 seeds are 1

(a) all round yellow **(b)** wrinkled yellow
(c) round green **(d)** all wrinkled green

9. The functional unit of the kidney is: 1

(a) nephron **(b)** neuron **(c)** villus **(d)** alveolus

10. The hormone responsible for the fight or flight response is: 1

(a) adrenaline **(b)** growth hormone **(c)** thyroxine **(d)** insulin

11. Sperms are produced in the 1

(a) testes **(b)** prostate **(c)** urethra **(d)** ureter

12. In human beings, the sex of the child is determined by 1

(a) environmental temperature
(b) the mother's diet
(c) the egg alone
(d) the type of sperm that fertilises the egg

13. The xylem in plants is responsible for: 1

(a) transport of amino acids **(b)** transport of food
(c) transport of oxygen **(d)** transport of water

14. The gap between two neurons is called: 1

(a) axon **(b)** myelin **(c)** dendrite **(d)** synapse

15. Multiple fission is seen in 1

(a) Hydra **(b)** Amoeba **(c)** Plasmodium **(d)** Planaria

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What are the differences between autotrophic nutrition and heterotrophic nutrition? 2

17. What is the difference between a reflex action and walking? 2

18. Why is variation useful for a species? 2

19. What is the difference between heredity and variation? 2

20. How is the small intestine designed to absorb digested food? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Why is the process of respiration considered an exothermic process? Write its equation. 3

22. Draw/describe the reflex arc for touching a hot object. 3

23. Describe the process of budding in Hydra. 3

24. A pure tall plant is crossed with a pure dwarf plant. Write the genotype and phenotype of F1 and the F2 ratio. 3

OR

Describe the working of the nephron in forming urine. 3

25. Name three animal hormones, the glands and the functions. 3

OR

Name the parts of a flower with their functions: 3
(a) anther
(b) stigma
(c) ovary.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the steps of photosynthesis: absorption of light, splitting of water, reduction of CO_2. What happens in desert plants at night? **5**

OR

Describe the structure of a neuron, with the path of a nerve impulse, and the synapse. **5**

27. Explain the different modes of asexual reproduction in single organisms with one example each. **5**

OR

Explain Mendel's monohybrid cross with tall and dwarf pea plants. Show the F1 and F2 generations with a Punnett square and the ratios. **5**

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND